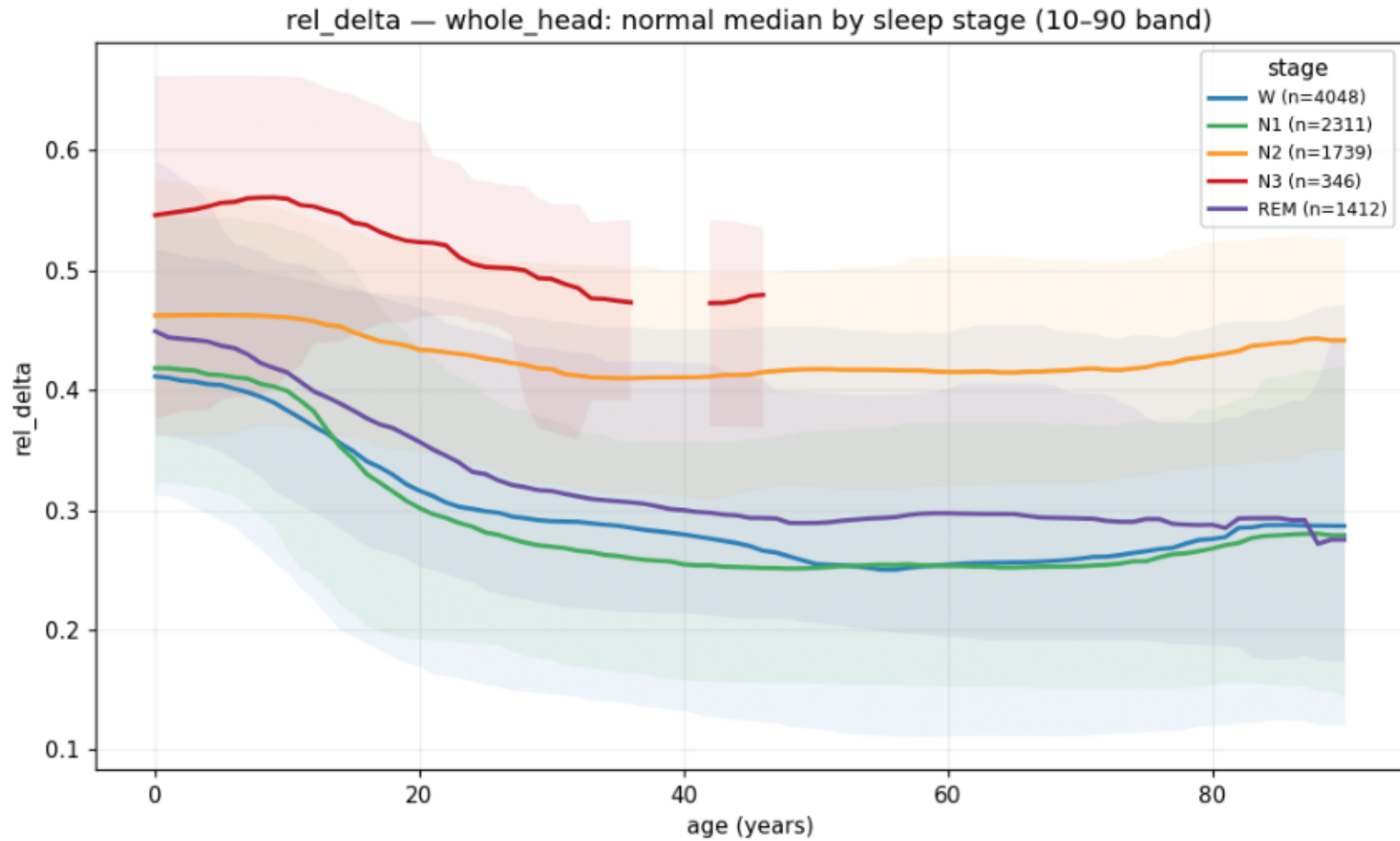


MORGOTH · SLOWING GROWTH CURVES (calibrated)

EEG slowing: growth curves, sleep-stage norms & gated report

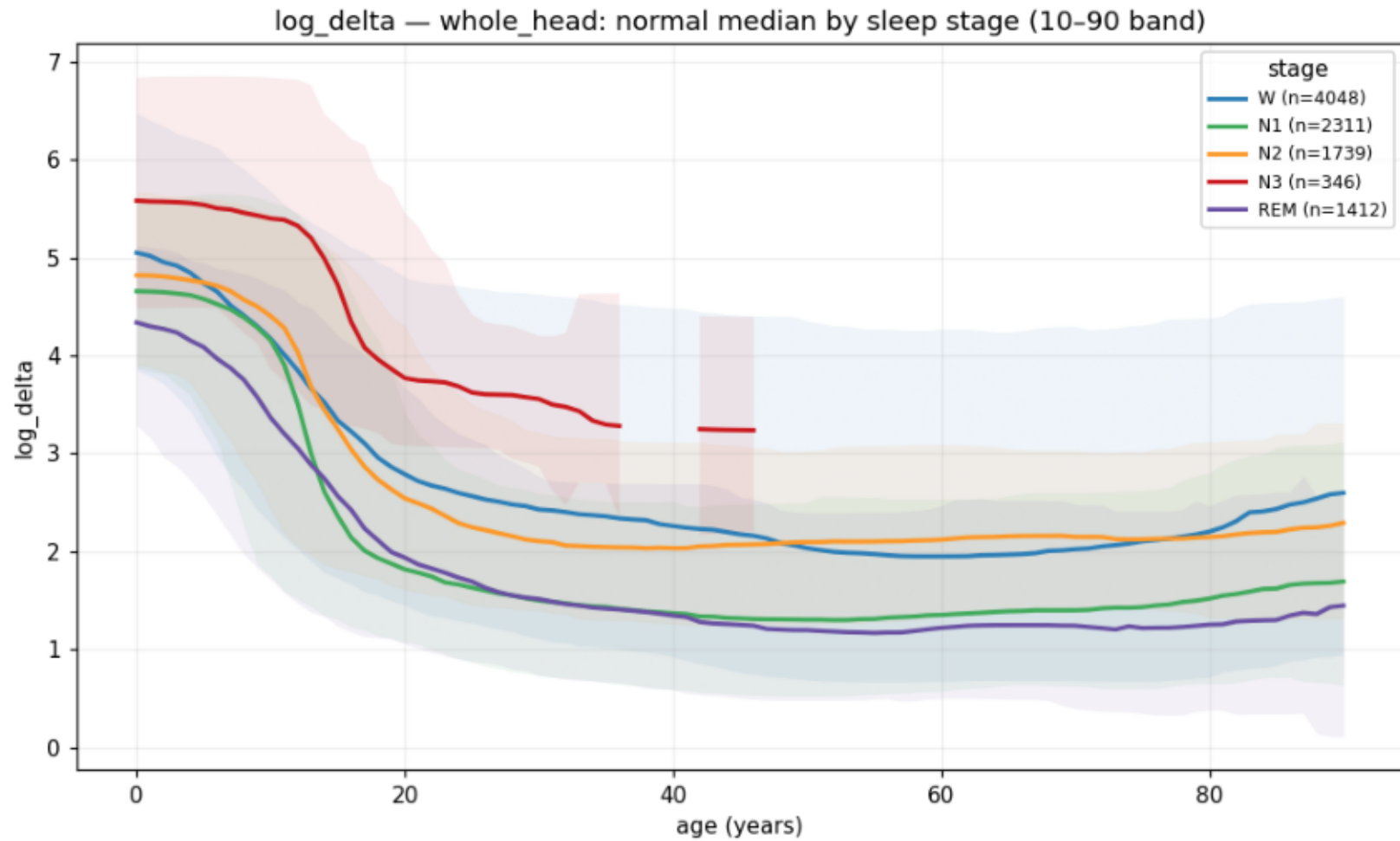
12,379 recordings. Delta 1-4 Hz calibrated (normal rel-delta 0.34). Morgoth gate (expert-calibrated) decides whether/focal-vs-generalized; normative deviation features add region/side/band/prevalence/persistence/sleep-stage. Validated vs clinical reports: region 0.92, side 0.79, band 0.74.

Relative delta by sleep stage (calibrated)



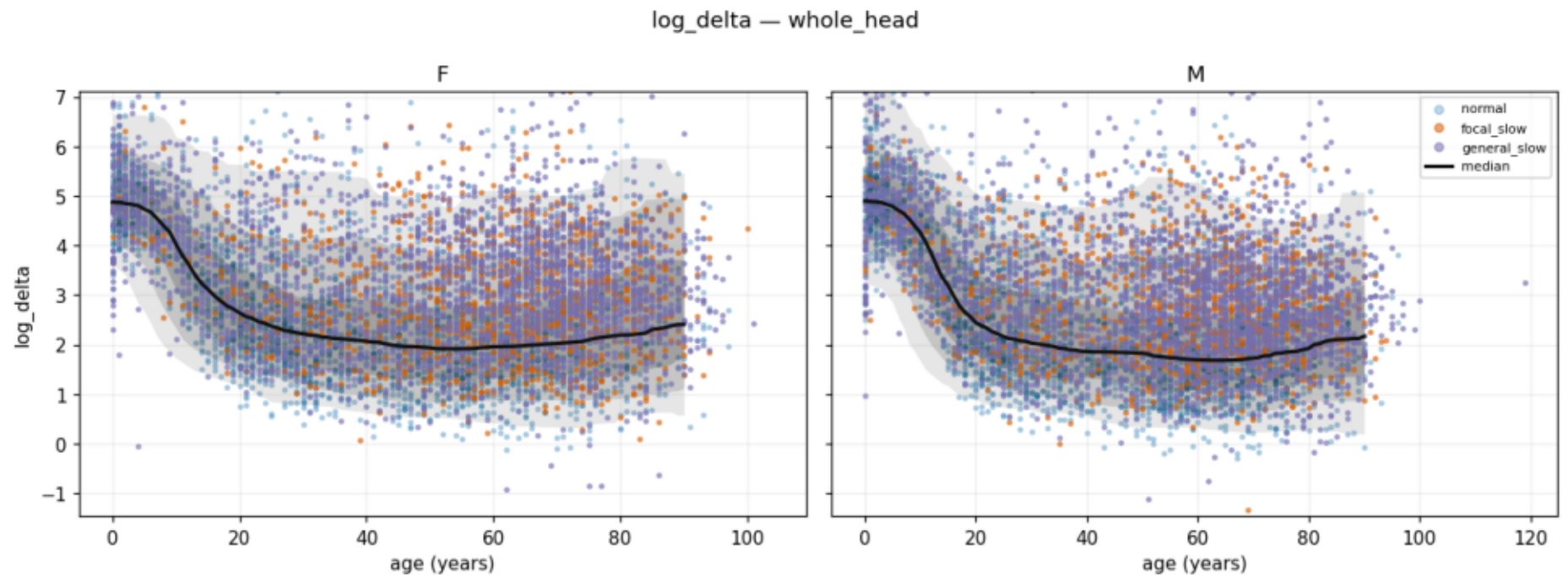
Normal delta rises $W \sim N1 < N2 < N3$; each segment scored vs its own stage's norm. rel_delta now physiological (~ 0.34 wake).

Absolute delta by sleep stage



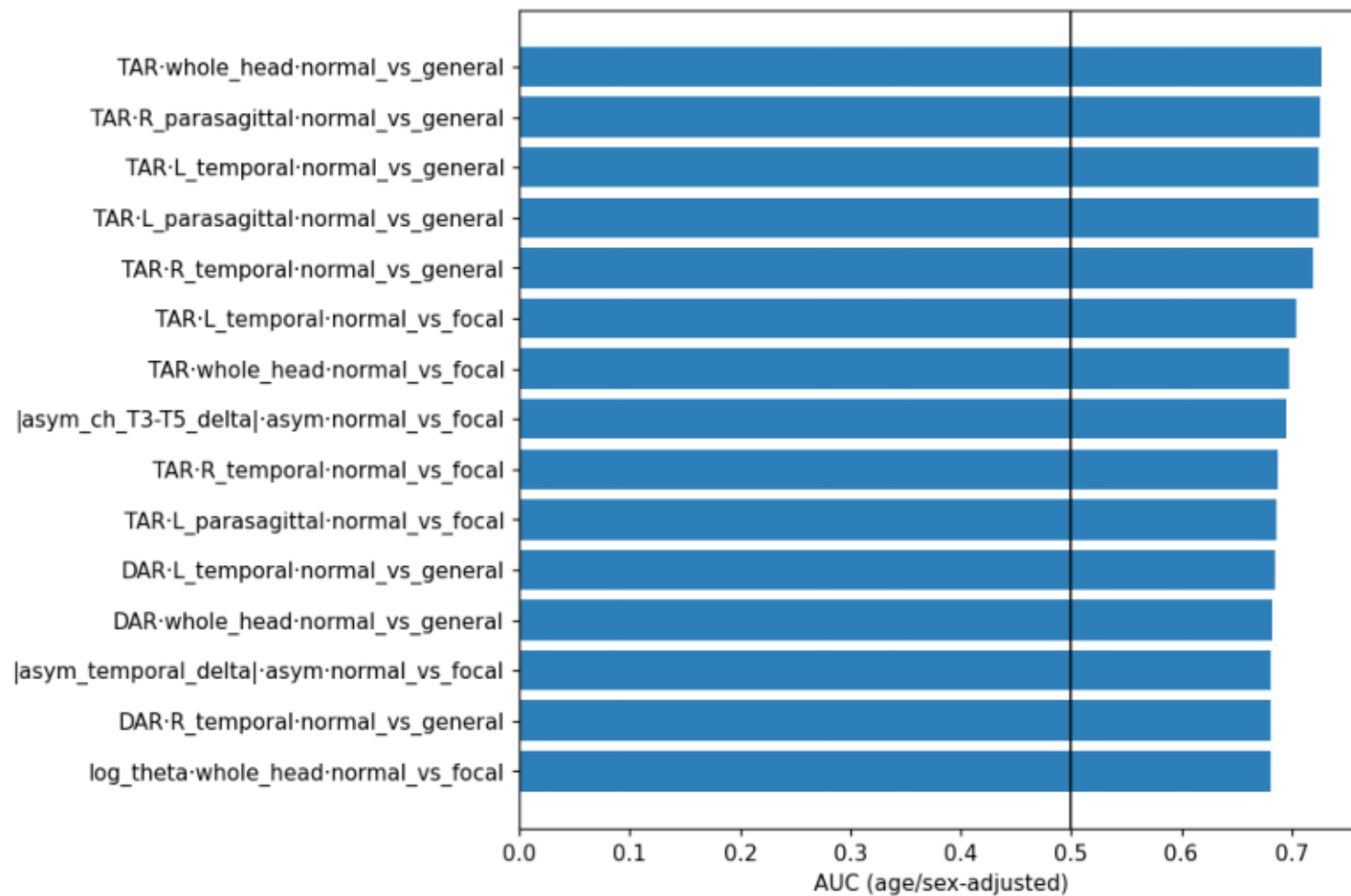
Deep sleep highest, wake lowest.

Absolute delta - development



Falls through childhood to a plateau ~30; slowing groups above the band.

Discriminative features (adjusted AUC)



TAR + per-channel homologous asymmetry lead.

Example gated report sentences

Focal-slowness

- (age 50 F, $P_{\text{slow}}=0.32$, focal) Frequent mild left temporal mixed theta/delta slowing — present in 48% of segments; peak 2.0 SD above age/stage-matched norms; longest run 3.8 min over 4 episodes.
- (age 38 F, $P_{\text{slow}}=0.88$, generalized) Frequent mild generalized mixed theta/delta slowing — present in 19% of segments; peak 1.3 SD above age/stage-matched norms; longest run 0.7 min over 6 episodes; present only during sleep; accentuated in N1.
- (age 71 M, $P_{\text{slow}}=0.53$, focal) Frequent mild left parasagittal mixed theta/delta slowing — present in 17% of segments; peak 2.3 SD above age/stage-matched norms; longest run 0.7 min over 4 episodes; accentuated in N3.
- (age 71 M, $P_{\text{slow}}=0.55$, generalized) Occasional mild generalized mixed theta/delta slowing — present in 7% of segments; peak 1.1 SD above age/stage-matched norms; longest run 0.5 min over 2 episodes.

Generalized-slowness

- (age 42 M, $P_{\text{slow}}=0.31$, generalized) Rare mild generalized mixed theta/delta slowing — peak 0.0 SD above age/stage-matched norms.
- (age 70 M, $P_{\text{slow}}=0.88$, generalized) Rare mild generalized mixed theta/delta slowing — peak 0.6 SD above age/stage-matched norms.
- (age 51 M, $P_{\text{slow}}=0.92$, generalized) Rare mild generalized mixed theta/delta slowing — peak 2.4 SD above age/stage-matched norms.
- (age 56 M, $P_{\text{slow}}=0.96$, generalized) Rare mild generalized mixed theta/delta slowing — peak 0.7 SD above age/stage-matched norms.